SYNOPSYS CHAMPIONSHIP PROJECT ABSTRACT

The Abstract is a required part of your project. Bring 1 copy to Project Check-In.

Your abstract should be written after you finish your research and experimentation and should include:

- Project number & project title
- Full name of every team member
- Purpose of your project
- Procedure
- Observations/data/results
- Conclusions

Type or print neatly using 10- or 12- point black font, single spaced. Your abstract should be 250 words or less and fit within the frame.

E13 Abstract
Project Title: A Convex Directional Spline Approach with Huber-Regularized Loss for Stable Deep Optimization

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Machine learning models power medical diagnosis, autonomous vehicles, and climate prediction, yet their accuracy is limited by static loss functions. Even a 1% accuracy gain could mean thousands of additional cancers detected early or millions saved in accident costs. Loss functions are formulas which guide a model to improve its predictions, and even the most widely used loss functions like Mean Squared Error (MSE) and Huber offer minimal mechanisms to adjust when data distributions shift.

This study proposes and evaluates a learnable Differing-Degree Spline loss function with Lagrangian dual variables that reshapes itself during training by learning optimal knot placements under monotonicity constraints, enabling it to adapt to data shifts in real-time. While prior work supports learnable loss parameters, this synthesis of Differing-Degree Splines with Lagrangian constraints for adaptive loss learning is novel.

The Lagrangian Differing-Degree Spline (LDDS) was evaluated under varying outlier contamination. Results showed average test improvements of 5-9% over MSE, and 4-7% over Huber across protein folding, diabetes, and image classification datasets. Evaluated on datasets with 40% outliers, LDDS achieved a 13.9% improvement while converging 11% faster than MSE. Further, dual variables enforced monotonicity, with violations converging to zero ($P = 0.012$ for MSE improvement; $P = 0.0011$ for constraint satisfaction).

These results suggest LDDS could result in higher accuracy under data distribution shifts. Moreover, it could cut training time by ~14% resulting in accelerated deployments of machine learning systems, reducing ~0.9 million metric tonnes of CO_2 caused by excess compute annually.